

Lightweight Hardware Architectures for the Piccolo Block Cipher in FPGA

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Abstract—The Piccolo block cipher is a lightweight block encryption for hardware use. Hardware devices are equipped with limited computation resources and small memory. In this paper, we propose an implementation to carry out through several trade-offs between area and speed. We implemented the Piccolo block cipher algorithm with 128-bit key in two different architectures on FPGA: the iterative and the 4-bit serial architectures. The proposed implementation was performed on Xilinx Spartan-3. The iterative implementation achieves 76% of resource utilization. This implementation takes 31 clock cycles to perform the encryption or decryption. So, it results in a throughput of 151.1 Mbps. The serial implementation was optimized in terms of area to reduce the cost. It achieves 54% of resource utilization and takes 496 clock cycles resulting in a throughput of 6.39 Mbps.

Keywords—lightweight block cipher, Piccolo block cipher, FPGA, Cryptography, Hardware architectures, low area, VHDL

I. INTRODUCTION

The born of new applications, such as Internet of Things IoT, Radio Frequency Identification RFID, and Wireless Sensor Network WSN, presents limited power and real-time constraints [1] [2]. Because of the limited resource of those applications, many problems arise and thus need to be solved. The surface and latency of an encryption system are major problems [3]. The use of classical encryption algorithms, such as AES, DES, 3DES and RSA, consume a large number of logical gates in order to provide optimal security. The performance of conventional encryption algorithms is not satisfactory and reliable as far as new applications are concerned because they require a lot of hardware resources and energy. So, it makes them not adapted for this type of limited resources applications. In recent years, many encryption algorithms, called lightweight encryption, with a high performance are developed. They have been designed to respect all the constraints of limited resource applications. Lightweight encryption algorithms such as LED [4], Present [5], Piccolo [6], Speck [7], and Simon [7] consume small area implementation of encryption algorithms.

Piccolo is a lightweight block cipher optimized for hardware implementation. It was presented for the first time at CHES 2011 [6]. According to Kyoji Shibutani et al, Piccolo has been developed with minimal hardware implementation, it is suitable for passive RFID tags. The architectural exploration for the Piccolo block cipher with serial and iterative implementation can be found in [6]. The

architectural design exploration has been implemented on ASIC. The first design is the serial architecture. It occupies 757.75 GE (gate equivalent), a latency of 528 clock cycles with a throughput of 12.12 Kbps. The second design is the iterative architecture. It consumes 1196.50 GE, a latency of 33 clock cycles with a throughput of 193.94 Kbps.

In this paper, we propose two different architectures, the iterative and the serial, for Piccolo block cipher based on the key length of 128-bits. These architectures are synthesized on Xilinx Spartan-3 XC3S500pq208-5 FPGA device. The iterative architecture utilizes 584 FPGA slices, has a latency of 31 clock cycles, runs at maximum clock frequency of 73.21 MHz, and provides a throughput of 151.1 Mbps. Our serial architecture uses 416 of FPGA slices, has a latency of 496 clock cycles, runs at a maximum clock frequency of 49.541 MHz, and provides a throughput of 6.392 Mbps.

The rest of the paper is organized as follows: an overview of the specification of the Piccolo algorithm is given in section II. Section III describes the proposed architectures for the Piccolo cipher iterative and serial architectures. Section IV provides experimental results and a discussion of the obtained results. In Section V, we conclude the paper.

II. DESCRIPTION OF THE PICCOLO ALGORITHM

In this section, a detailed description of the different versions of the Piccolo block cipher was given. Piccolo is an iterative block cipher based on a Feistel Network (FN) structure that was proposed by Kyoji Shibutani et al in 2011 [6]. It can use a block size message of 64-bits with two different key sizes: 80-bits and 128-bits. This algorithm has 25 rounds when 80-bit key size is used and 31 rounds for the 128-bits. An additional two rounds are performed. A mix keys WK_i function is performed at the top and the bottom of the algorithm, which results in total rounds of 27 for Piccolo-80 and 33 for Piccolo-128. The specification for the encryption function of Piccolo is presented in Alg. 1.

Algorithm 1 Encryption procedure for Piccolo as described in [6] where $r = \{25, 31\}$ for $\{\text{Piccolo-80, Piccolo-128}\}$, respectively.

Input: A 64-bit plaintext P, n-bit secret key K.
Output: A 64-bit ciphertext C.
S $\leftarrow$ P
 $WK_0 \leftarrow$ GenerateWkey (K)
 $WK_{31} \leftarrow$ GenerateWkey (K)
S $\leftarrow$ AddWkey (S, WK_0)

For $i = 1$ to r do
 $rK_i \leftarrow \text{GenerateRoundkey}(K)$
 $S \leftarrow \text{AddFunction}(S)$
 $S \leftarrow \text{AddRoundkey}(S, rK_i)$
 $S \leftarrow \text{RP}(S)$

End for

$C \leftarrow \text{AddWkey}(S, WK_{31})$

Return C

Both the ciphertext and plaintext have the same block size of 64-bits. The plaintext is transformed into a ciphertext after n rounds of the Piccolo algorithm. The architecture of the Piccolo encryption algorithm is shown in Figure 1. More details about Piccolo can be found in [6].

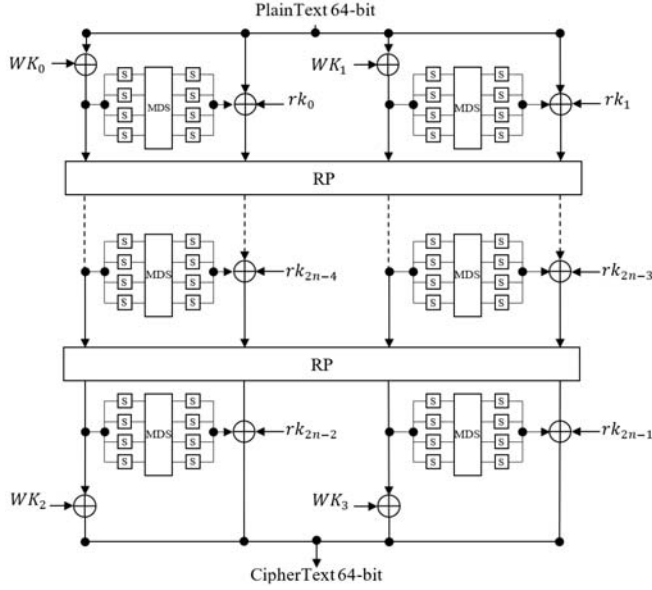


Fig. 1. The Encryption procedure of Piccolo

III. HARDWARE IMPLEMENTATION OF THE PICCOLO ALGORITHM

In this section, a detailed description of the proposed implementation of Piccolo blocks cipher in serial architecture on 4-bits and iterative architecture was presented. For all the implementations in this paper, the same Input/Output interface was fixed between the encryption and the external environment for both architectures. The Top-level interface is illustrated in Figure 2.

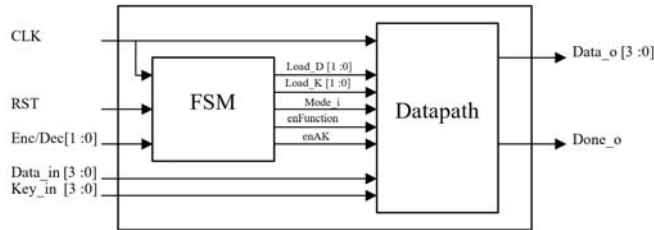


Fig. 2. The Top-level Input/Output interface of Piccolo

Datapath is used to encrypt or decrypt the input data. It is composed of a round function and registers to load the input data and key of Piccolo. The finite state machine (FSM) generates five control signals, which are load_D, load_K, enFunction, enAK and Mode_i. It is used to control the key generation process and encryption/decryption mode.

The signals enFunction and enAK are used in serial architecture to enable the two operations: AddRoundKey and AddFunction. Load_D and load_K are signals that control the data and key loading. Mode_i is used to enable the encrypt or decrypt round function.

A. Iterative Architecture

In this work, we proposed the use of 4-bits Input/Output data ports (4-bits for the plaintext, 4-bits for the key and 4-bits for the output). The data is stored in a 64-bit register and the key in a 128-bit register (for the key in 128-bits). In the proposed architecture, every round takes one clock cycle to perform the round function.

Processing a 64-bit plaintext block with key in a 128-bits takes 32 clock cycles to load data and key, 31 latency cycles for encryption and 16 cycles to generate the output. This results in a 79 clock cycles to encrypt the plaintext. The proposed iterative architecture is illustrated in Figure 3.

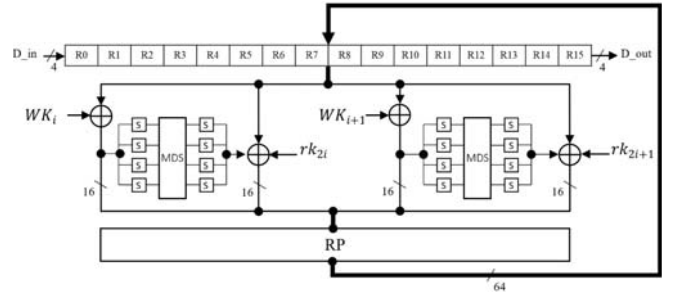


Fig. 3. Proposed iterative architecture of Piccolo

This type of architecture uses two Feistel schemas. Each Feistel schema uses 32-bits Input/Output data, it is composed of function F and XOR gates. XOR gates are used to add the output function F and round key rK_i with 16-bits of data. The F function uses 16-bits Input/Output of data; it is composed of two S-Box layers separated by diffusion matrix MDS. The output of the first S-Box layer is multiplied by a diffusion matrix M which is presented in (1).

$$V * M = (V_0 \ V_1 \ V_2 \ V_3) * \begin{pmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 2V_0 \oplus V_1 \oplus V_2 \oplus 3V_3 \\ 3V_0 \oplus 2V_1 \oplus V_2 \oplus V_3 \\ V_0 \oplus 3V_1 \oplus 2V_2 \oplus V_3 \\ V_0 \oplus V_1 \oplus 3V_2 \oplus 2V_3 \end{pmatrix} \quad (1)$$

The hardware architecture of the F function is present in figure 4. The cost area of two F functions in a round function consume 24 XOR gates, 16 S-Box blocks and 16 multipliers.

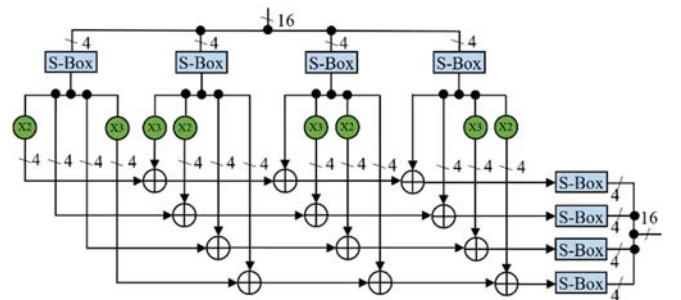


Fig. 4. Proposed hardware architecture of the F function

The subkey operation WK_i is performed in the first and last rounds. It is additional the subkey WK_i with 32-bits of data. The Subkey generator of Piccolo is implemented by using multiplexers to select the appropriate round key rK_i or subkey WK_i for each round.

B. Serial Architecture

To save area and energy, we propose a serial implementation of the Piccolo algorithm. In effect, the round function is optimized in order to reduce the implementation area and power consumption compared to the iterative architecture. The round function architecture is composed of registers, multiplexers and XOR gates. The proposed round function for the serial implementation is shown in Figure 5.

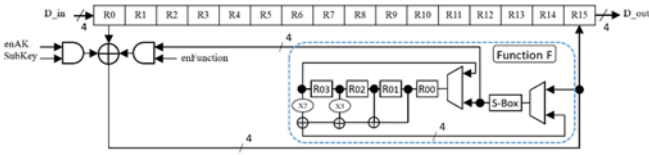


Fig. 5. Proposed round function for serial implementation

The optimized F function uses less area compared to that in the iterative architecture. In the proposed round function one S-Box block, four registers, three XOR gates and two multiplexers 2/1 are used. The 16-bits data in four registers $R_{00}R_{01}R_{02}R_{03}$ are multiplied by vector A that is suitable for a compact serial architecture. Vector A is equivalent to the matrix M that is used in an iterative implementation. To get the same results the registers $R_{00}R_{01}R_{02}R_{03}$ are shifted to the left every clock cycle and the multiplication is repeated until all the registers are shifted. An example of the proposed multiplication by the vector A is presented in (2).

$$V * A = (R_{00} R_{01} R_{02} R_{03}) * \begin{pmatrix} 2 \\ 1 \\ 1 \\ 1 \end{pmatrix} = (2R_{00} \oplus R_{01} \oplus R_{02} \oplus 3R_{03}) \quad (2)$$

Processing a 64-bit plaintext block with key in a 128-bit takes 32 clock cycles to load data and key, 16 clock cycles for every round, and 16 cycles to generate the output, which gives a 544 clock cycles to encrypt the plaintext. The proposed serial architecture is illustrated in Figure 6.

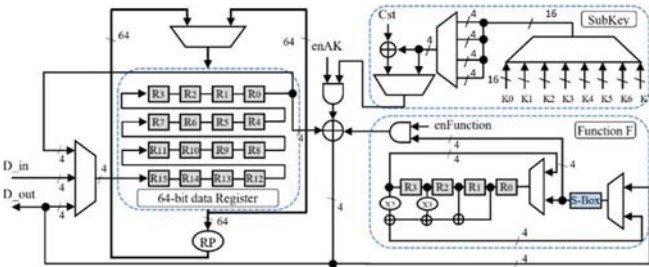


Fig. 6. Proposed serial architecture of Piccolo

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, we detail the proposed implementation on the Spartan-3 XC3S50pq208-5 FPGA device. Table I presents the performance and resource usage results for the

iterative and serial architectures of Piccolo in FPGAs devices.

TABLE I. RESOURCE USAGE AND PERFORMANCE RESULTS FOR THE PROPOSED IMPLEMENTATION OF PICCOLO ON THE SPARTAN-3 FPGA DEVICES

Architectures	Iterative		Serial 4-bits	
	Enc	Enc/Dec	Enc	Enc/Dec
SLC	397	584	265	416
FF	207	211	260	285
LUT	757	1132	442	729
Freq MAX (MHz)	81.82	73.21	45.85	49.54
Latency (clock cycles)	31	31	496	496
Throughput (Mbps)	168.9	151.1	5.92	6.39
Thr/SLC (Mbps/Slice)	0.495	0.294	0.022	0.015

Here, the proposed iterative architecture uses 584 FPGA slices. It has a latency of 31 clock cycles and runs at a maximum frequency of 73.21 MHz, which gives a throughput of 151.1 Mbps. For the proposed serial architecture, it occupies 416 of FPGA slices. It has a latency of 496 clock cycles and runs at a maximum frequency of 49.54 MHz, which gives a throughput of 6.39 Mbps.

TABLE II. COMPARISON OF PERFORMANCE VERSUS EXISTENCE WORK

	Architecture	Latency	Throughput
Proposed	Iterative	31	151.1 Mbps
	Serial	496	6.39 Mbps
Piccolo [5]	Iterative	33	193.9 Kbps
	Serial	528	12.12 Kbps

Table II presents a comparison between the original and the proposed implementations. The existing architectures are given in [6]; it has been performed in ASIC. For this reason, to compare the architectures with the existing designs, the selected metrics are latency and throughput of the Piccolo algorithm. Throughput is computed using formula (3).

$$throughput = \frac{Max\ freq * Size\ of\ block}{Latency} \quad (3)$$

A comparison of latency is shown in Figure 7. Both the proposed architectures require lower latency than architectures in [6]. The proposed iterative architecture takes less latency by two clock cycles to run hardware implementation of Piccolo block cipher. In serial architecture, the proposed implementation is lower than the implementation in [6] by 32 clock cycles. Also, both proposed architectures achieve high throughput (Mbps).

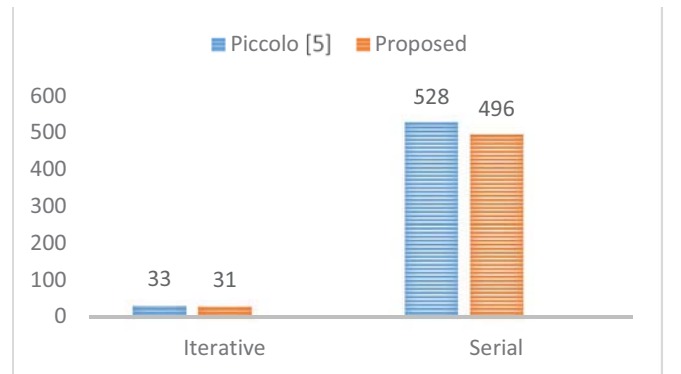


Fig. 7. Comparison of latency for different implementations of Piccolo

TABLE III. FPGA IMPLEMENTATION RESULT OF OTHER BLOCK CIPHERS

<i>Block Ciphers</i>	<i>Devices</i>	<i>SLC</i>	<i>Latency</i>	<i>F(MHz)</i>	<i>Thr(Mbps)</i>
PRESENT[8]	Spartan3 XC3S50-5	117	256	114.8	28.7
LED[12]	Spartan-3 XC3S50-5	227	48	87.63	116.54
PRESENT[9]	Spartan-3 XC3S400-5	202	32	254	508
SEA[13]	Virtex-II XC2V4000	424	-	145	-
XTEA[11]	Spartan3 XC3S50-5	254	112	62.6	35.77
PRINT[10]	Spartan3 XC3S700A N-5	210	48	147.73	147.7

Table III presents the existing FPGA implementation results of other block ciphers for use in performance and resource comparisons with our Piccolo implementation results. The proposed iterative architecture of Piccolo achieve high throughput (Mbps) compared to the other implementation results in Table III.

The proposed serial architecture of Piccolo encryption uses 265 of FPGA slices. It is average results compared to results in Table III.

V. CONCLUSION

In this paper, we proposed two efficient architectures for Piccolo block cipher with key sizes of 128-bits. The proposed architectures have been modeled in VHDL language and synthesized on the Xilinx Spartan-3 XC3S50pq208-5 FPGA device. In comparison to other existing implementations, the proposed architectures show low latency and provide high throughput, which makes them suitable for applications with high throughput. To the best of our knowledge, we have obtained the first optimized implementations and the first FPGA results for the Piccolo block cipher. As far as future work is concerned, we plan to deploy the proposed implementation for IoT application.

REFERENCES

- [1] L. Su Mi, H. Young Ju, L. Dong Hoon and L. Jong In, "Efficient Authentication for Low-Cost RFID Systems," International Conference on Computational Science and Its Applications. Springer, Berlin, Heidelberg, Vol. 3480, pp. 619-627, 2005.
- [2] Q. Jing, A. Vasilakos, J. Wan, J. Lu and D. Qiu, "Security of the Internet of Things: perspectives and challenges," Wireless Networks, vol. 20, pp.2481-2501, 2014.
- [3] A. Juels, "Minimalist cryptography for low-cost RFID tags," International conference on security in communication networks. Springer, Berlin, Heidelberg, Vol. 3352, pp. 149-164, 2004.
- [4] J. Guo, T. Peyrin, A. Poschmann and M. Robshaw, "The LED block cipher," International Workshop on Cryptographic Hardware and Embedded Systems. Springer, Berlin, Heidelberg, Vol. 6917, pp. 326-341, 2011.
- [5] A. Bogdanov, L. knudsen, G. Leander, C. Paar, A. Poschmann, M. Robshaw, Y. Seurin and C. Vikkelsoe, "PRESENT: An ultra-lightweight block cipher," International workshop on cryptographic hardware and embedded systems. Springer, Berlin, Heidelberg, Vol. 4727, pp.450-466, 2007.
- [6] K. Shibutani, T. Isobe, H. Hiwatari, A. Mitsuda, T. Akishita and T. Shirai, "Piccolo: an ultra-lightweight blockcipher," International Workshop on Cryptographic Hardware and Embedded Systems. Springer, Berlin, Heidelberg, Vol. 6917, pp.342-357, 2011.
- [7] R. Beaulieu, S. Treatman-Clark, D. Shors, B. Weeks, J. Smith and L. Wingers, "The SIMON and SPECK lightweight block ciphers,"

IEEE, 2015 52nd ACM/EDAC/IEEE Design Automation Conference (DAC), pp.1-6, 2015.

- [8] P. Yalla, J.P. Kaps. Lightweight Cryptography for FPGAs. In Recon_gurable Computing and FPGAs, 2009. ReConFig'09. International Conference on, pages 225-230. IEEE, 2009.
- [9] A.Y. Poschmann. LIGHTWEIGHT CRYPTOGRAPHY: Cryptographic Engineering for a Pervasive World. In Ph.D. Thesis. 2009.
- [10] T. Okabe, "Efficient FPGA Implementations of PRINTCIPHER", International Journal of Emerging Technologies and Innovative Research, Vol 3, Issue 4, pp.76-85, 2016.
- [11] J.P. Kaps. "Chai-Tea, Cryptographic Hardware Implementations of xTEA". In Progress in Cryptology INDOCRYPT 2008, pages 363-375. Springer, 2008.
- [12] N. Nalla Anandakumar, T. Peyrin, A. Poschmann, "A Very Compact FPGA Implementation of LED and PHOTON", Lecture Notes in Computer Science, Vol 8885, pp.304-321, INDOCRYPT 2014.
- [13] F. Mace, F.X. Standaert, J.J. Quisquater. "FPGA Implementation(s) of a Scalable Encryption Algorithm". Very Large Scale Integration (VLSI) Systems, IEEE Transactions on, 16(2):212-216, 2007.